

Tangents



- 1 Find the gradient of the tangent to $f(x) = x^3 + 5x$ at the point $(2, 18)$.
- 2 Consider the function $y = \frac{1}{3}x^3 + \frac{1}{6}x^6 + 2x$.
 - a Find $\frac{dy}{dx}$.
 - b Evaluate $\frac{dy}{dx}$ when $x = -7$.
- 3 Find the gradient of $f(x) = \frac{11}{x} + \frac{10}{x^2}$ at the point $\left(4, \frac{27}{8}\right)$.
- 4 Consider the parabola $f(x) = x^2 + 3x - 10$.
 - a Find the x -intercepts.
 - b Find the gradient of the tangent at the positive x -intercept.
- 5 Consider the function $y = 4x^2 - 5x + 2$.
 - a Find $\frac{dy}{dx}$.
 - b Find the value of x at which the tangent to the parabola is parallel to the x -axis.
- 6 Consider the tangent to the curve $f(x) = -x^3$ at the point $(2, -8)$.
 - a Find the gradient of the function $f(x) = -x^3$ at the point.
 - b Hence, find the equation of the tangent to the curve at the point.
- 7 Consider the tangent to the curve $f(x) = 5\sqrt{x}$ at the point $\left(\frac{1}{9}, \frac{5}{3}\right)$.
 - a Find the gradient of the function $f(x) = 5\sqrt{x}$ at the point.
 - b Hence, find the equation of the tangent to the curve at the point.
- 8 Consider the tangent to the curve $f(x) = 6\sqrt{x}$ at the point $(4, 12)$.
 - a Find the gradient of the function $f(x) = 6\sqrt{x}$ at the point.
 - b Hence, find the equation of the tangent to the curve at the point.

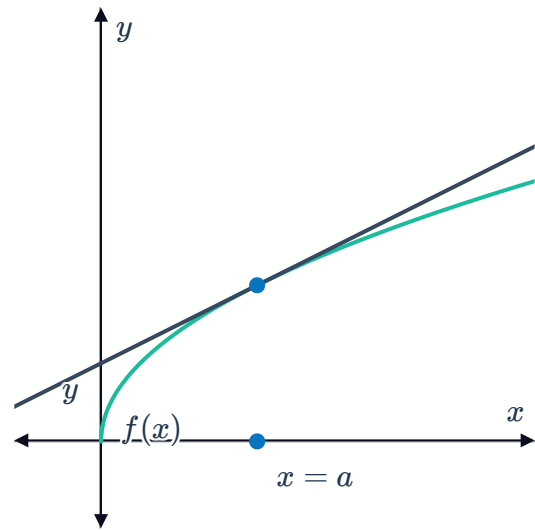
- 9 Consider the tangent to the curve $f(x) = \frac{2}{x^3}$ at $x = -2$.
- Find the gradient of the function $f(x) = \frac{2}{x^3}$ at $x = -2$.
 - Find the y -coordinate of the point of intersection between the tangent line and the curve.
 - Hence, find the equation of the tangent to the curve at $x = -2$.
- 10 Consider the tangent to the curve $f(x) = -2x^2 + 8x + 2$ at $x = 1$.
- Find the y -coordinate of the point of intersection between the tangent line and the curve.
 - Find the equation of the tangent to the curve at $x = 1$.
- 11 Consider the tangent to the curve $f(x) = 3x^3$ at $x = 2$.
- Describe the steps required to find the equation of the tangent line at $x = 2$.
 - Hence, find the equation of the tangent to the curve.
- 12 Consider the function $f(x) = \frac{9x+4}{3x}$.
- Find the y -coordinate of the point on the curve at $x = -1$.
 - Hence, find the equation of the tangent to the curve at $x = -1$.
- 13 Consider the function $g(x) = \frac{8x^7 - 6x^6 + 4x^5 + 7}{2x^2}$.
- Find the derivative of the function.
 - Find the equation in of the tangent to the curve at $x = 1$.
- 14 Find the equation of the tangent to $f(x) = x^2 + x$ at the point $(2, 6)$.
- 15 Find the equation of the tangent to $f(x) = (3x - 1)(2x - 5)$ at the point $(2, -5)$.
- 16 Consider the function $f(x) = x^3 - 6x^2$. Find the coordinates of the points on the curve where the gradient is 495.
- 17 Consider the function $f(x) = x^3 + 3x^2 - 19x + 2$. Find the coordinates of the points on the curve where the gradient is 5.
- 18 Consider the function $f(x) = 2x^2 - 216\sqrt{x}$. Find the coordinates of the point on the curve



- 19 Consider the function $f(x) = \frac{4x^3}{3} + \frac{5x^2}{2} - 3x + 7$. Find the x -coordinates of the points on the curve whose tangent is parallel to the line $y = 3x + 7$.
- 20 At point M , the equation of the tangent to the curve $y = x^2$ is given by $y = 4x - 4$. Find the coordinates of M .
- 21 At point M , the equation of the tangent to the curve $y = x^3$ is given by $y = 12x - 16$. Find the coordinates of M .
- 22 Find the equation of the tangent to the parabola $y = 2x^2 + 8x - 5$ at the point where the gradient is 0.
- 23 Consider the curve given by the function $f(x) = x^2 - 4x + 2$.
- a Find the gradient of the tangent to the curve at the point $(3, -1)$.
 - b Find the coordinates of the vertex of the parabola $f(x)$.
 - c Sketch the graph of the curve and the tangent at the point $(3, -1)$.
- 24 Consider the curve given by the function $f(x) = x^2 - 1$.
- a Find the gradient of the tangent to the curve at the point $(1, 0)$.
 - b Sketch the graph of the curve and the tangent at the point $(1, 0)$.
- 25 Consider the curve $y = x^3 - x^2$ and the line $4x - y = 11$.
- a Find the x -coordinates of the points on the curve at which the tangents are perpendicular to the line $4x - y = 11$.
 - b Find the equation of the tangent to the curve at each of the x -coordinates found in part (a).
- 26 The curve $f(x) = k\sqrt{x} - 5x$ has a gradient of 0 at $x = 16$. Find k .

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In the figure, the straight line $y = \frac{1}{10}x + b$ tangent to the graph of $f(x) = 6\sqrt{x}$ at $x = a$.



- a** Find a . **b** Find b .

28 $5x + y + 2 = 0$ is the tangent line to the curve $y = x^2 + bx + c$ at the point $(9, -47)$.

- a** Find the derivative $\frac{dy}{dx}$ of $y = x^2 + bx + c$.
b State the gradient of the tangent to the curve at $x = 9$.
c Find the value of b .
d Find the value of c .

29 The curve $y = ax^3 + bx^2 + 2x - 17$ has a gradient of 58 at the point $(2, 31)$.

- a** Use the given information to write two equations expressing b in terms of a .
b Hence, find the value of a and b .

30 The graph of $y = ax^3 + bx^2 + cx + d$ has the following properties:

- It intersects the y -axis at $y = -28$, where the tangent is parallel to the x -axis.
- It intersects the x -axis at $(2, 0)$, where it has a gradient of 36.

- a** Use the information given in the first point above to find the value of c and d .
b Use the information given in the second point above to write two equations expressing b in terms of a .
c Hence, find the value of a and b .

31 From an external point $(3, 2)$, two tangents L_1 and L_2 are drawn to the curve $y = x^2 - 6$.

- a** Let the gradient of a tangent line to the curve be m . Find the two possible values of m that correspond to the gradients of L_1 and L_2 .
b If L_1 is the tangent with the larger gradient, find the equation of L_1 .
c Find the equation of L_2 .



Normals

32 Consider the function $y = 10x^5 - 2x^4 + 8x^3 - 6x^2 + 4x - 164$.

- a** Find $\frac{dy}{dx}$.
- b** Find the gradient of the tangent to the curve at the point $(3, 2278)$.
- c** Find the gradient of the normal to the curve at the point $(3, 2278)$.

33 The gradient of the curve $f(x)$ at $x = 3$ is 2.

- a** Find the gradient of the tangent line at $x = 3$.
- b** Find the gradient of the normal line at $x = 3$.

34 Consider the curve $f(x) = x^2 + 8x + 15$ at the point $(4, 63)$.

- a** Find $f'(x)$.
- b** Find the gradient of the tangent to the curve at the point.
- c** Find the equation of the tangent to the curve at the point.
- d** Find the gradient of the normal to the curve at the point.
- e** Find the equation of the normal to the curve at the point.

35 Consider the curve $f(x) = 4x + \frac{64}{x}$ at the point $(4, 32)$.

- a** Find the equation of the tangent to the curve at the point.
- b** Find the equation of the normal to the curve at the point.

36 $-4x + y + 1 = 0$ is the normal to the curve $y = x^2 + bx + c$ at the point $(-8, -32)$.

- a** Find the derivative $\frac{dy}{dx}$ of $y = x^2 + bx + c$.
- b** State the gradient of the normal to the curve at $x = -8$.
- c** Find the value of b .
- d** Find the value of c .



Applications

- 37** Consider the curve $f(x) = 36 - x^2$ at the point $P(3, 27)$.
- a** Find $f'(x)$.
 - b** Find the gradient of the tangent to the curve at the point.
 - c** Find the equation of the tangent to the curve at the point.
 - d** Find the gradient of the normal to the curve at the point.
 - e** Find the equation of the normal to the curve at the point.
 - f** The tangent line at $(3, 27)$ meets the x -axis at point A . Find the x -coordinate of A .
 - g** The normal line at $(3, 27)$ meets the x -axis at point B . Find the x -coordinate of B .
 - h** Find the length of AB .
 - i** Find the exact area of $\triangle ABP$.
- 38** The normal to the curve $y = x(x - 5)^2$ at the point $A(6, 6)$ cuts the x -axis at B .
- a** Find the gradient function.
 - b** Find the gradient of the tangent at A .
 - c** Find the gradient of the normal at A .
 - d** Hence, find the x -coordinate of point B .
 - e** Find the area of triangle whose vertices are the origin, point A and point B .